

Aluminum Cylinder Simulation

Geometry: Aluminum cylinder with a diameter 2 inches and length 10 inches.

Boundary Conditions: Temperatures of 300°F and 70°F were applied to the two ends respectively.

Objective: Use FEA to compute steady-state temperature distribution along the axis.

FEA Tool: ANSYS Workbench (Steady-State Thermal Analysis).

Method:

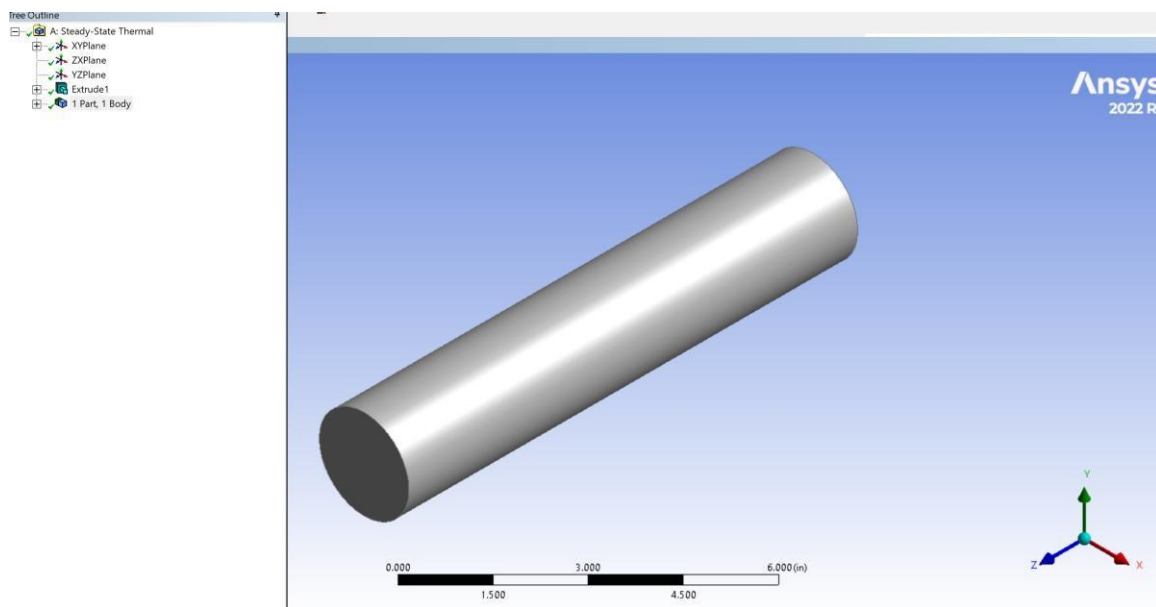
3D geometry was created and meshed.

Boundary conditions were applied at the ends.

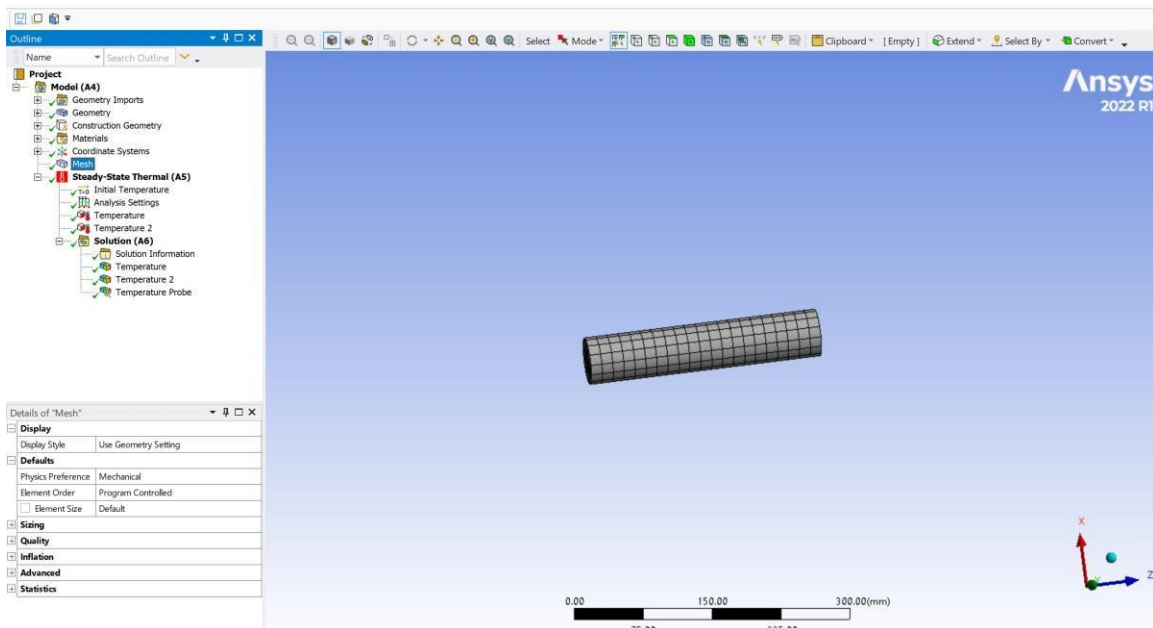
A path was created along the axis, and temperature results were extracted at intervals of 2 inches.

The results confirmed a linear drop in temperature from 300°F to 70°F across the length.

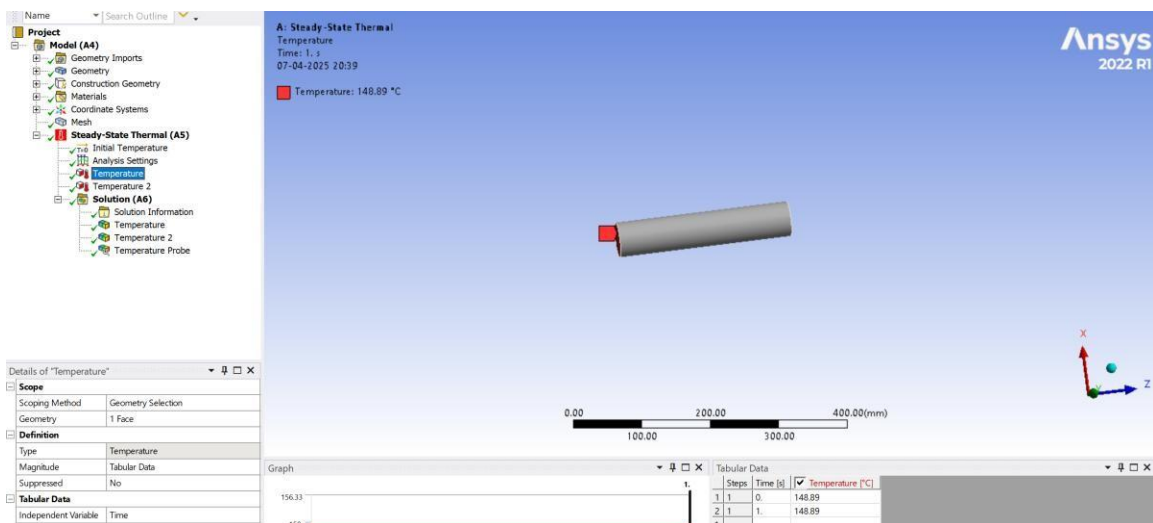
Model:

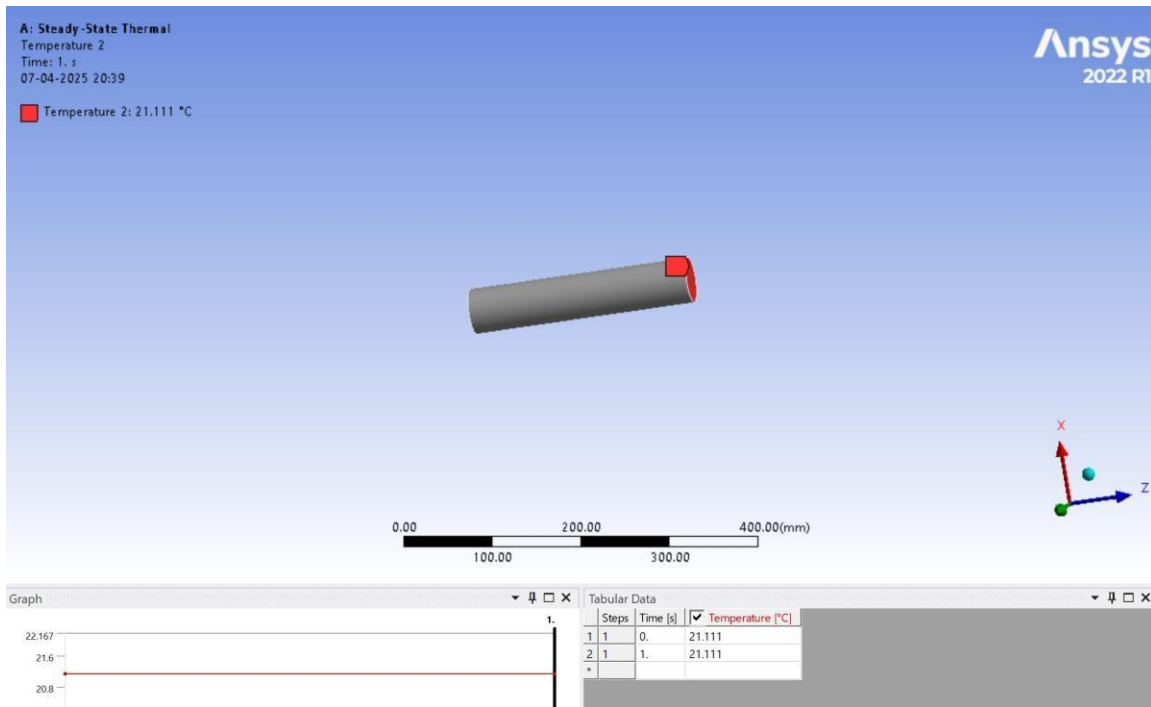


Mesh:

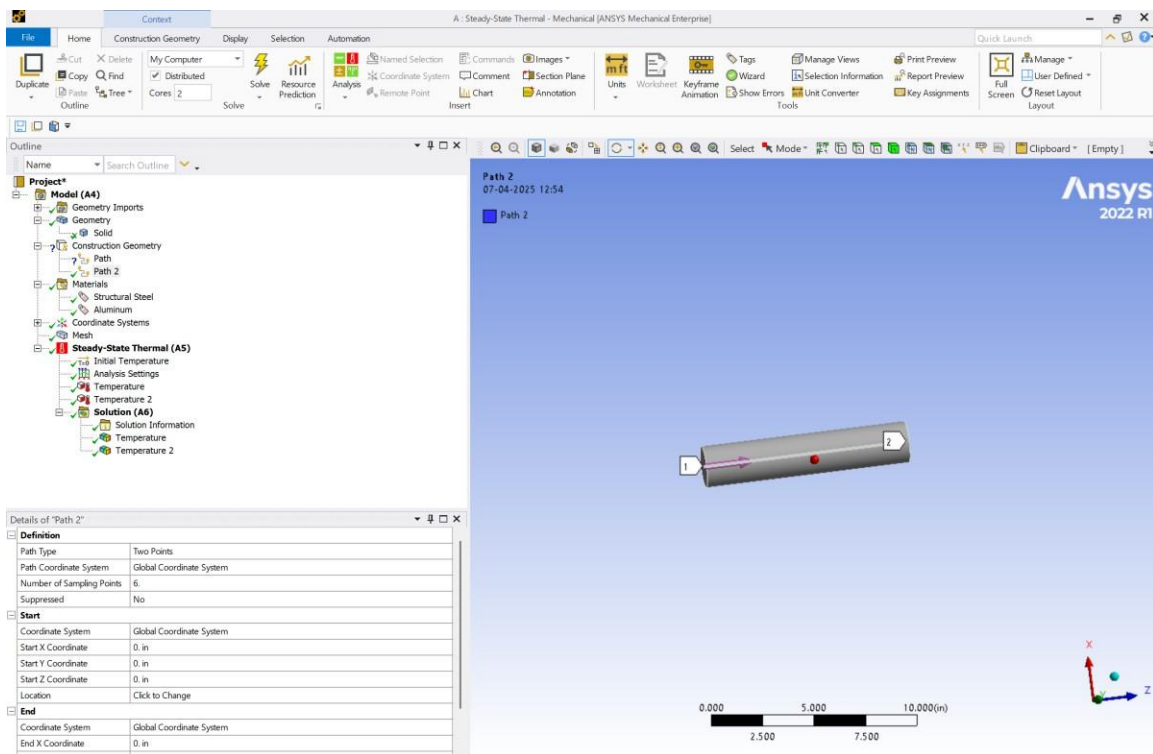


Temperature:

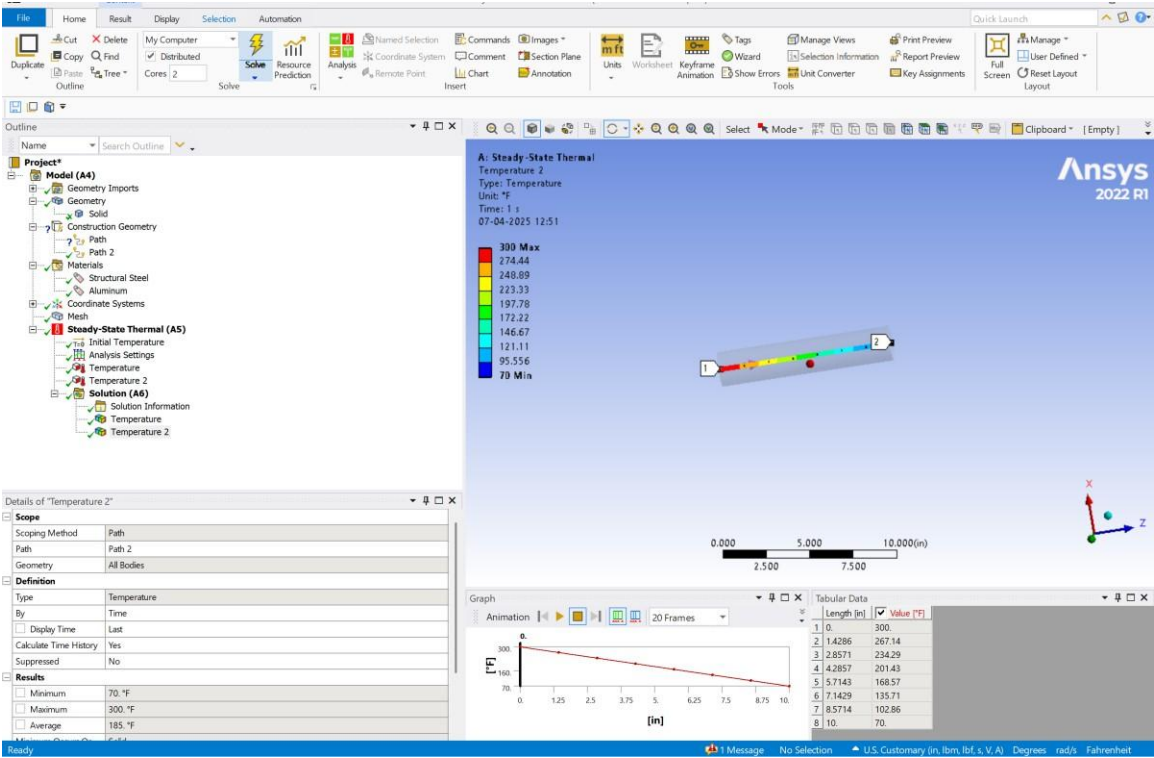
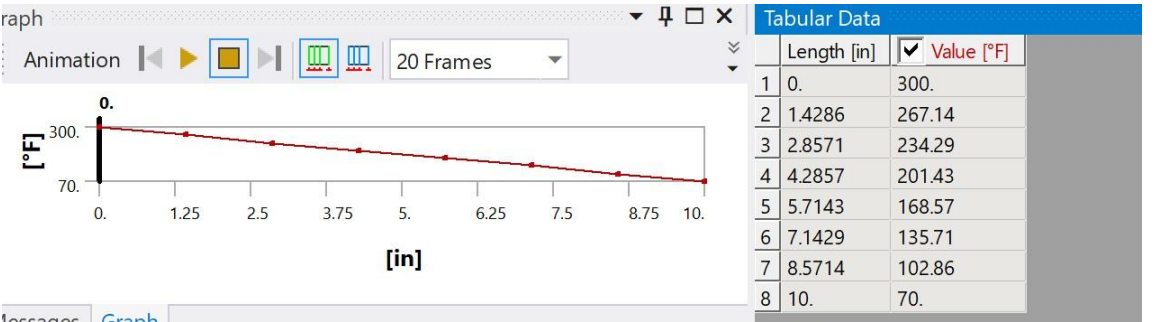




Path:



Result:



Aluminum Block Simulation

Geometry: Aluminum block with dimensions 20 in x 10 in x 10 in.

Boundary Conditions: Temperatures were applied at all 8 corner vertices as follows:

- T1: 700°F
- T2: 70°F
- T3: 30°F
- T4 :600°F
- T5 :900°F
- T6 :200°F
- T7 :500°F
- T8: 70°F

Objective: Determine temperature distribution inside the block using steady-state FEA.

Method:

Block was modeled in Design Modeler and meshed in Mechanical.

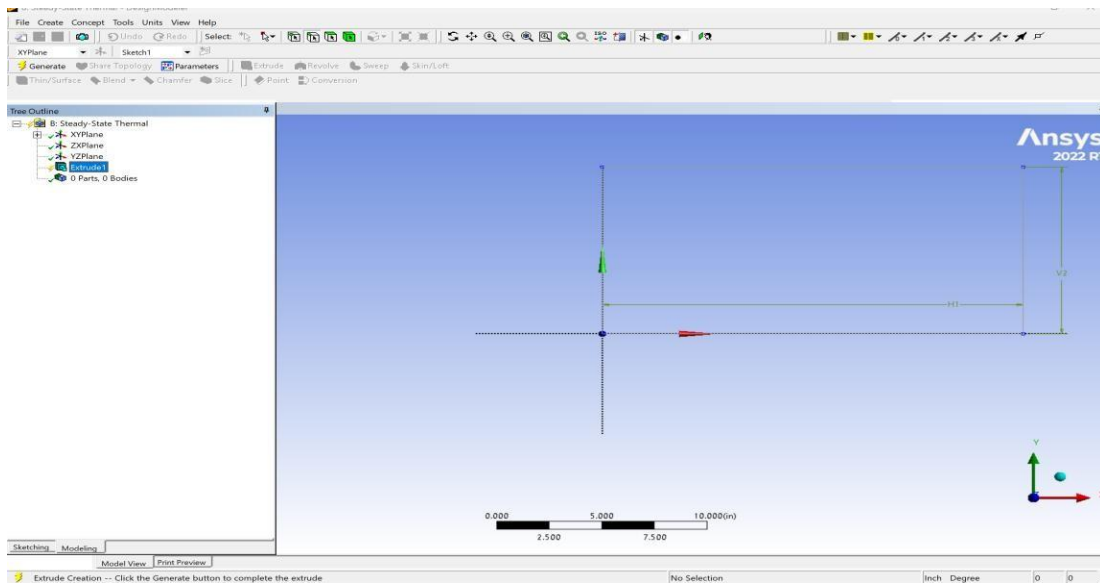
Temperatures applied to each vertex.

Named selections were created for each face.

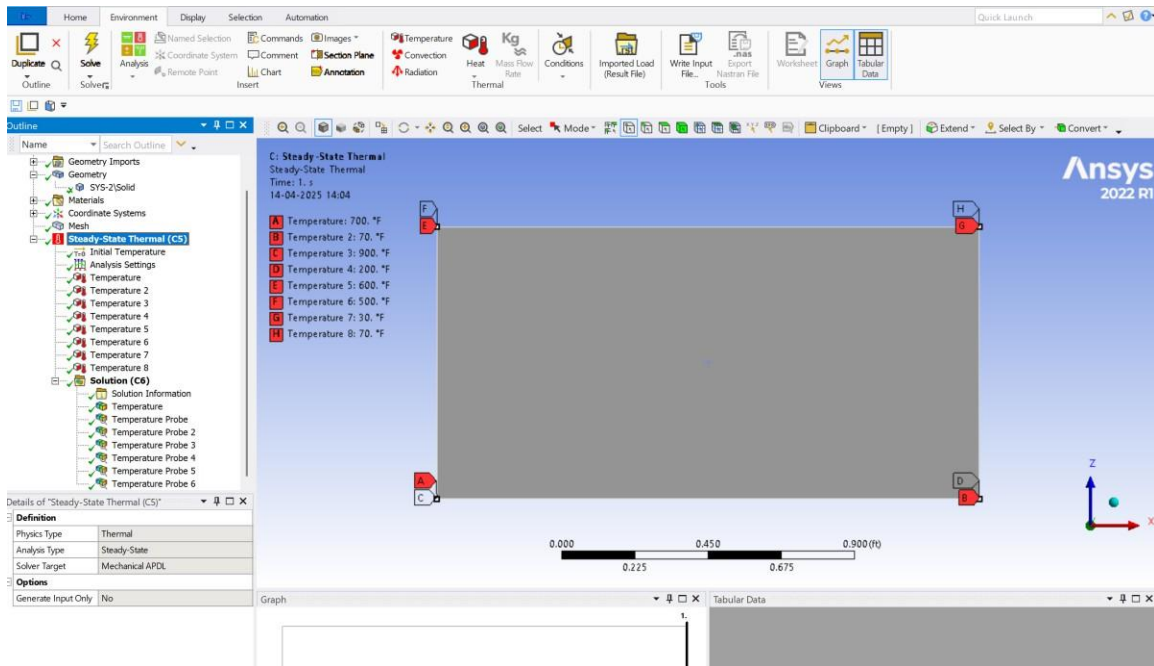
Average temperature values were extracted at the center of each face.

Used either face-scope temperature or a central path with a single point for exact values

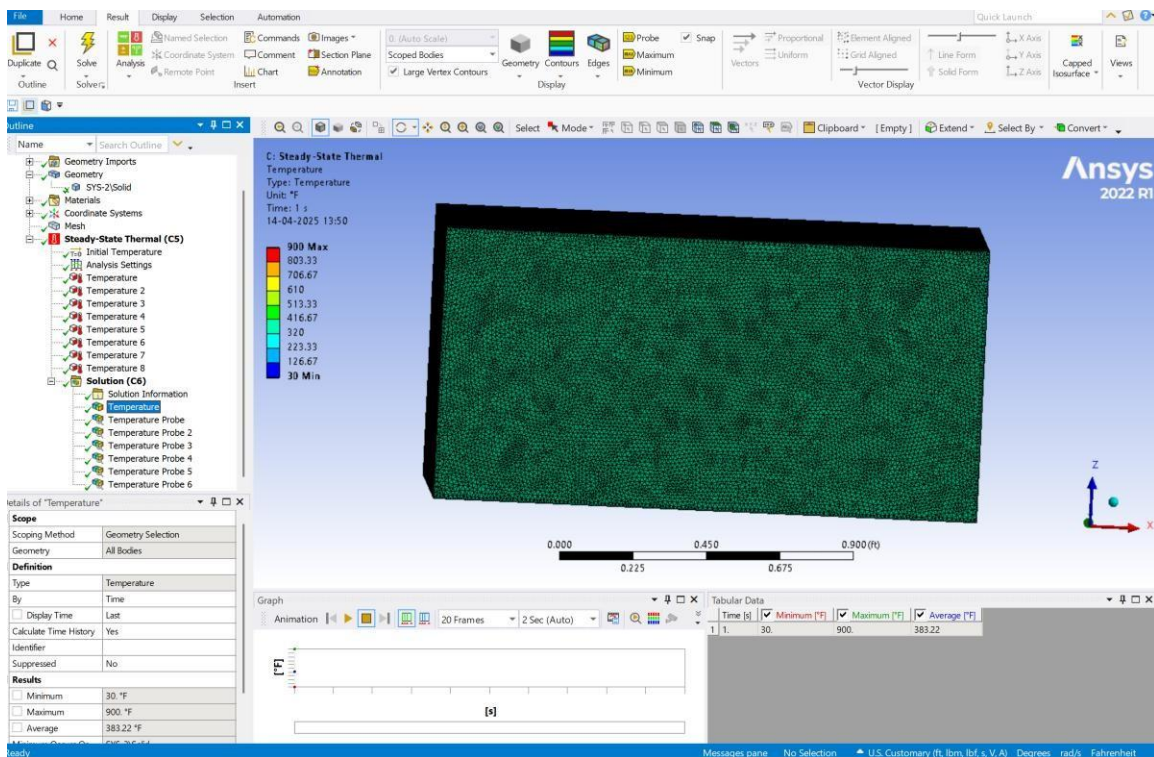
Geometry:

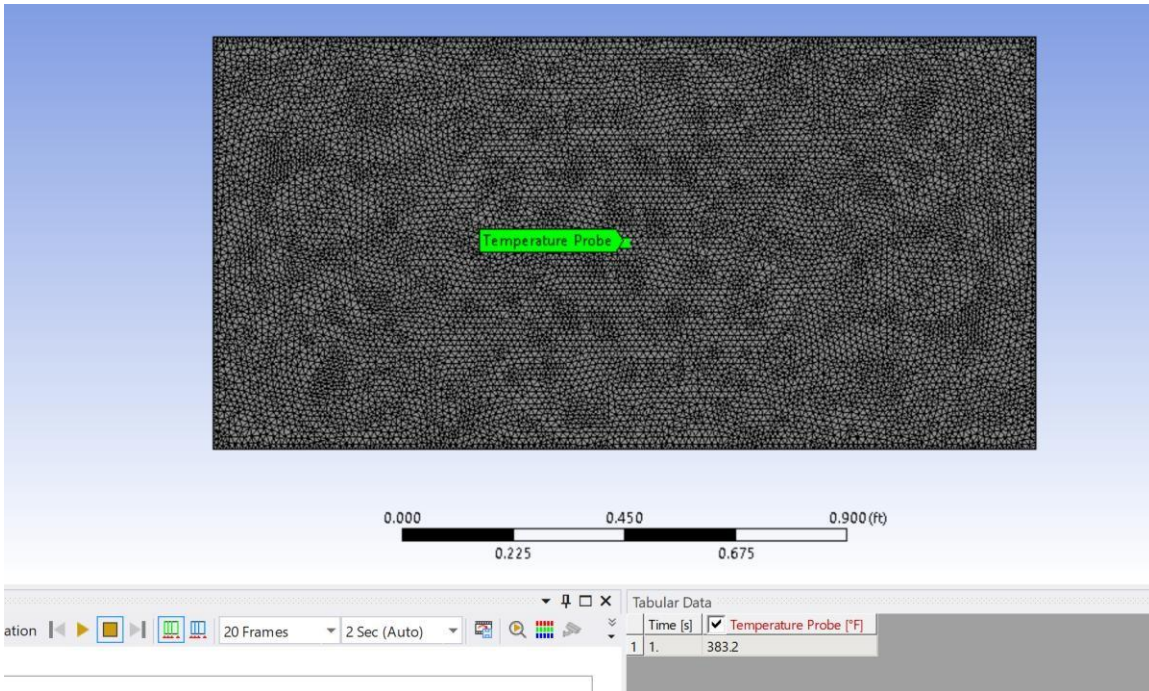


Temperature:

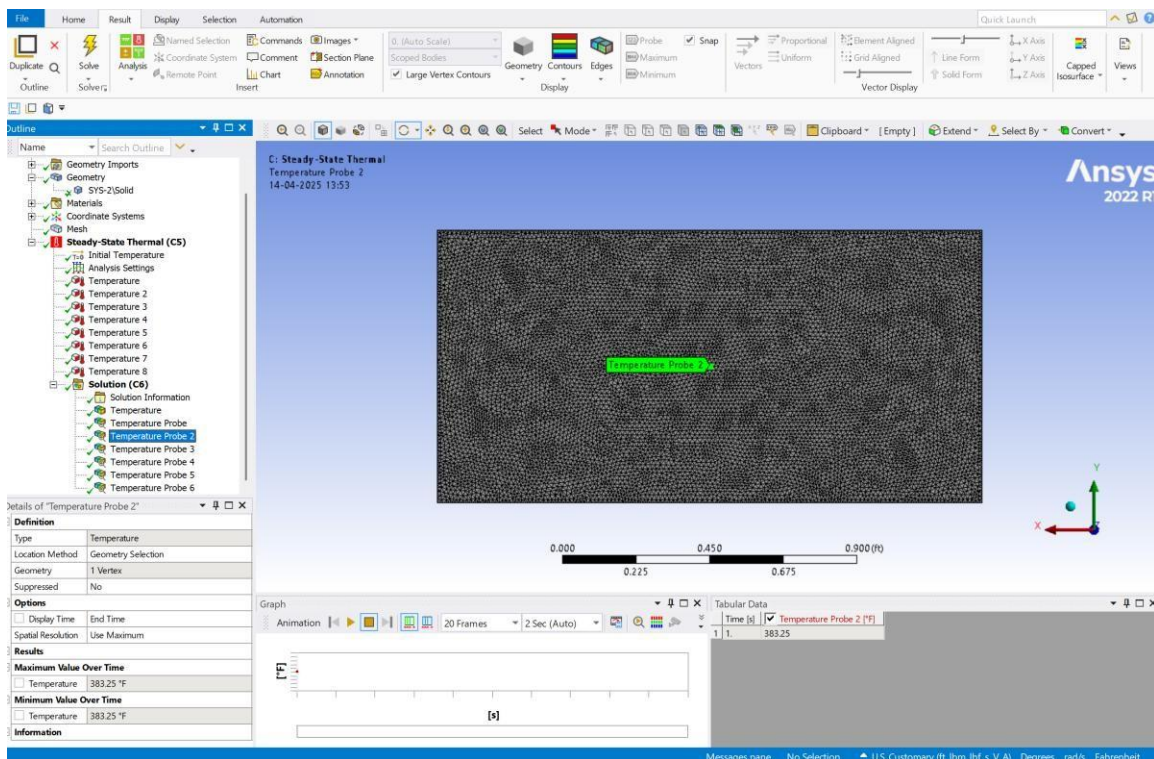


Results:

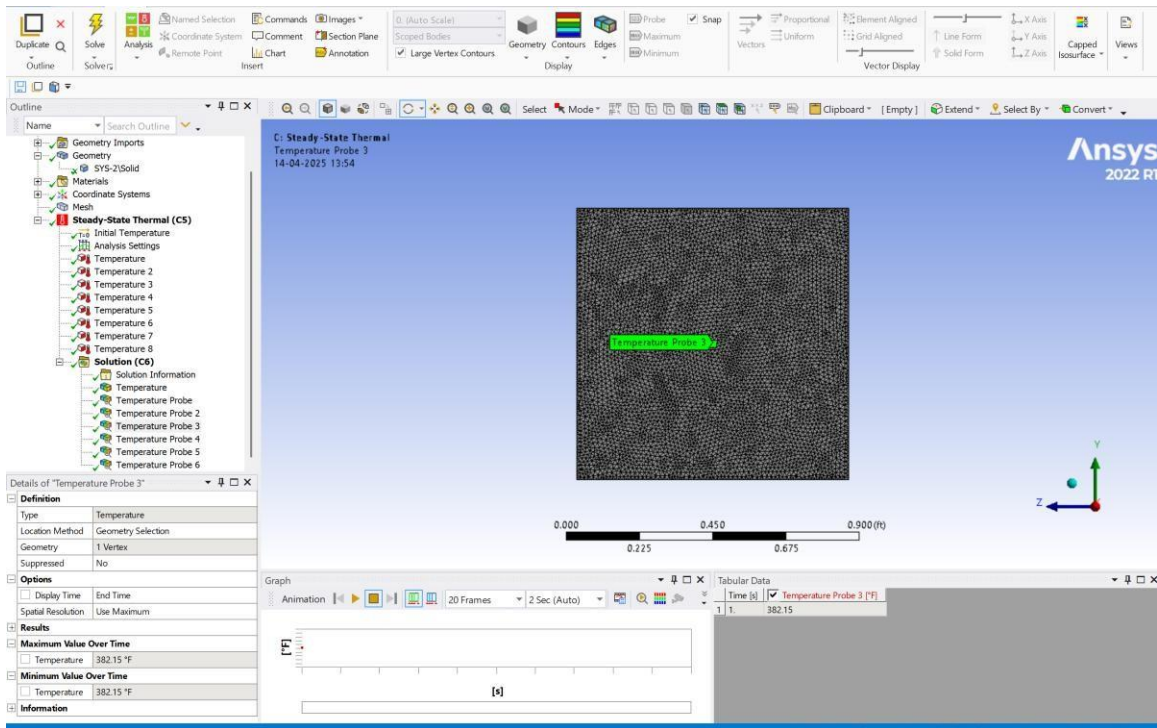




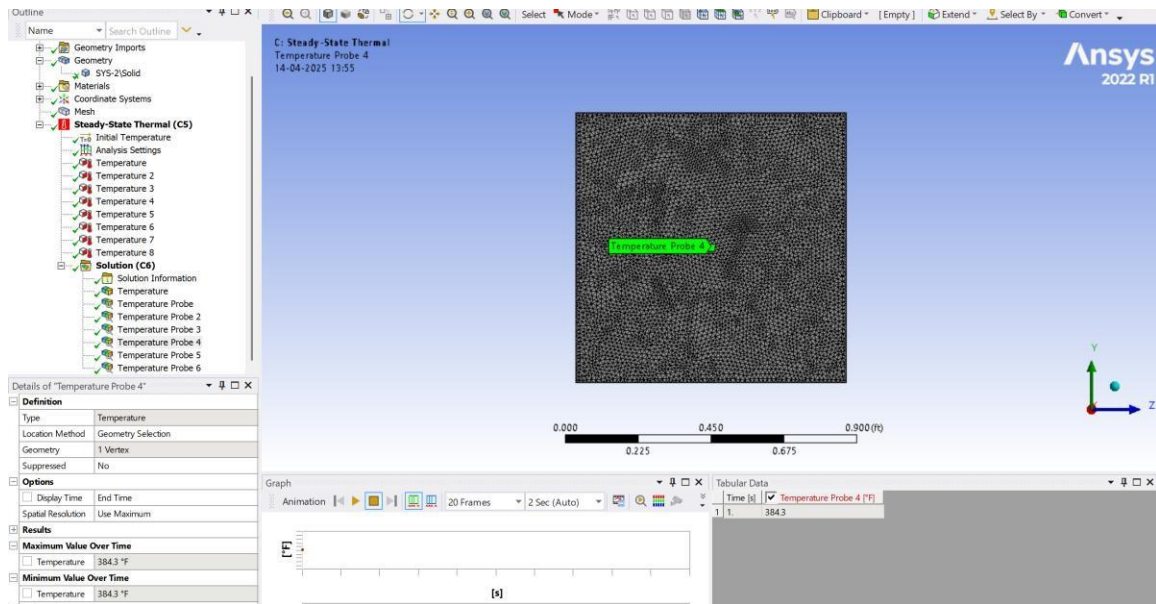
Front View



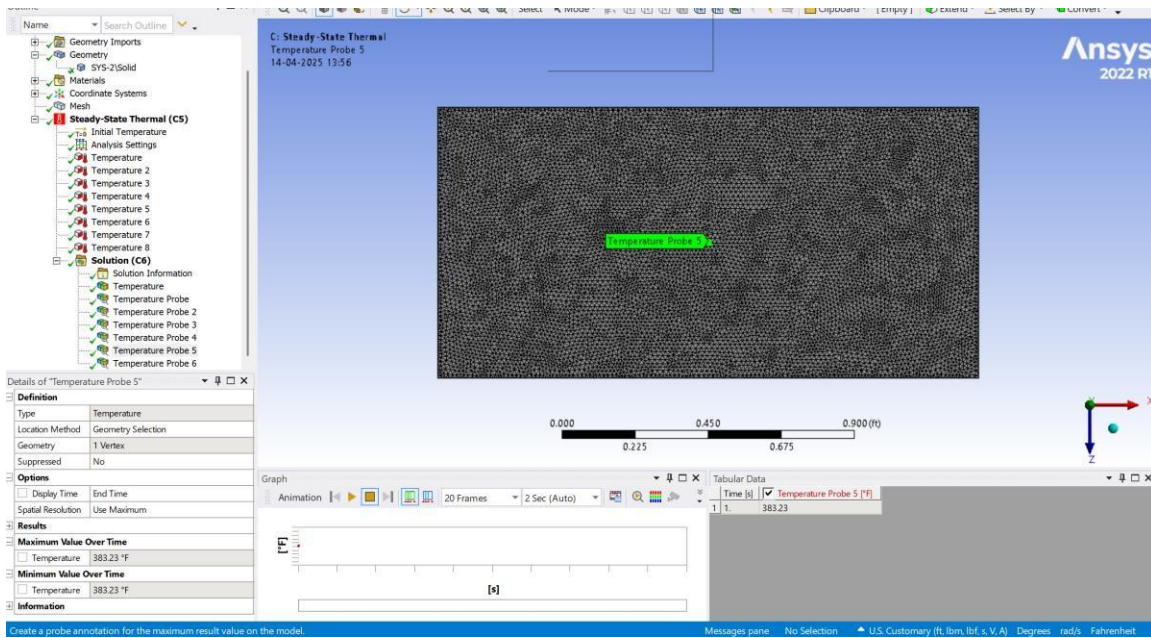
Back View



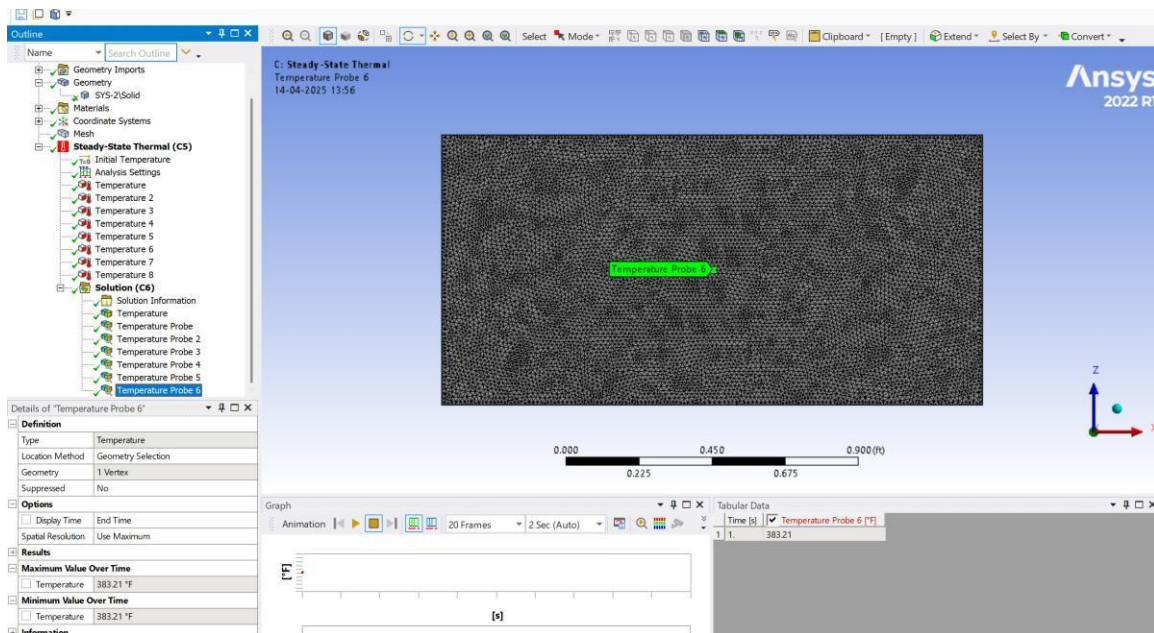
Right View



Left View



Top View



Bottom View